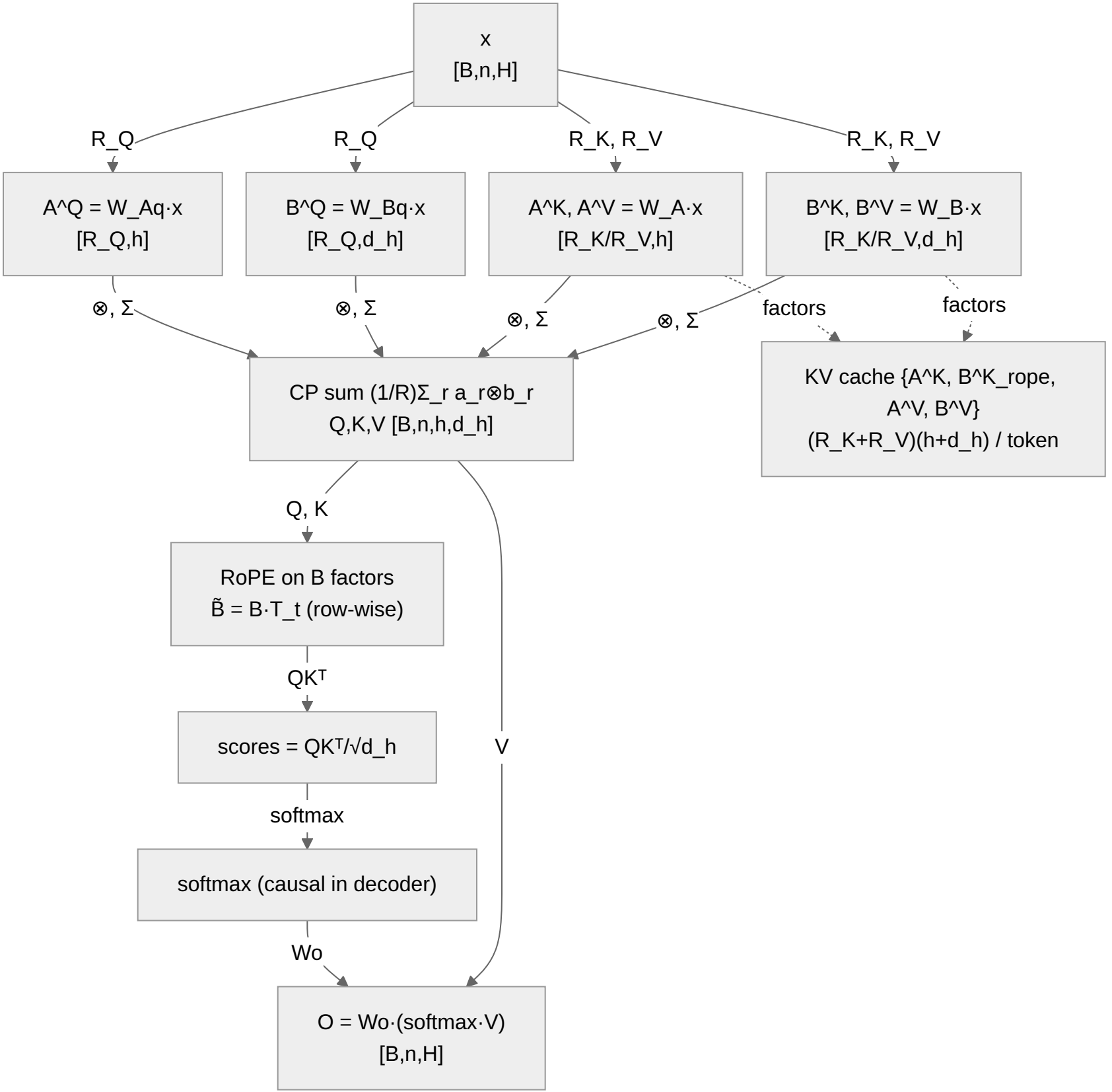
x
[B,n,H]
R_Q
R_Q
R_K, R_V
R_K, R_V
A^Q = W_Aq·x
[R_Q,h]
B^Q = W_Bq·x
[R_Q,d_h]
A^K, A^V = W_A·x
[R_K/R_V,h]
B^K, B^V = W_B·x
[R_K/R_V,d_h]
⊗, Σ
⊗, Σ
⊗, Σ
⊗, Σ
factors
factors
CP sum (1/R)Σ_r a_r⊗b_r
Q,K,V [B,n,h,d_h]
KV cache {A^K, B^K_rope,
A^V, B^V}
(R_K+R_V)(h+d_h) / token
Q, K
RoPE on B factors
B̃ = B·T_t (row-wise)
QKᵀ
scores = QKᵀ/√d_h
V
softmax
softmax (causal in decoder)
Wo
O = Wo·(softmax·V)
[B,n,H]